

Rigid-Body Transformations in Robotics

Lecture Summary & Formula Quick-Reference Guide

Course: COMS W4733 Computational Aspects of Robotics — Instructor: Tony Dear, Ph.D.

1 Fundamental Definitions & Frames

- **World Space ($\mathcal{W}$):** Modeled as $\mathbb{R}^2$ or $\mathbb{R}^3$.
- **Points vs. Vectors:** Points $p \in \mathcal{W}$ depend on the reference frame's position and orientation. Free vectors (e.g., linear velocities) have fixed magnitude and direction.
- **Frames:** Typically consists of one fixed *world frame* and one or more moving *body frames* attached to the robot.
- **Rigid-Body Transformation:** A mapping $h : \mathcal{W} \rightarrow \mathcal{W}$ preserving relative distances and orientations (excluding reflections/mirror images) between all points on the body.

2 Translations & Rotations

2.1 Translations

- **Formula:** $h(x, y, z) = (x + x_t, y + y_t, z + z_t)$.
- **Duality:** Translating a body by (x_t, y_t, z_t) is equivalent to inverse translation of the reference frame by (x_t, y_t, z_t) .

2.2 2D Rotations

- **2D Rotation Matrix** (Counterclockwise angle θ):

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

- **Frame Axis Interpretation:** Columns of $R(\theta) = [x_1 \ y_1]$ are the body frame axes expressed in the world frame.
- **Properties of Rotation Matrices:**
 - **Orthogonality:** $R(\theta)^{-1} = R(\theta)^T = R(-\theta)$ (inverse is clockwise rotation).
 - **Determinant:** $\det(R(\theta)) = 1$ (preserves length, size, and vector magnitude).
 - **Commutativity (2D):** $R(\theta_1 + \theta_2) = R(\theta_1)R(\theta_2) = R(\theta_2)R(\theta_1)$.

2.3 3D Rotations

- **Principal Axis Rotations:**

$$R_z(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}, \quad R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \gamma & -\sin \gamma \\ 0 & \sin \gamma & \cos \gamma \end{bmatrix}$$

- **Properties:** Orthogonal ($R^{-1} = R^T$) and $\det(R) = 1$.
- **Right-Hand Rule:** Thumb along rotation axis $\mathbf{v}$; curled fingers indicate positive (counterclockwise) rotation in the normal plane.
- **Non-Commutativity (3D):** 3D rotations about *different* axes do **not** commute ($R_y(\phi)R_z(\theta) \neq R_z(\theta)R_y(\phi)$).

3 Homogeneous Transformations (T)

Rigid-body motion $r = p + Rq$ is *affine*. Homogeneous coordinates express it as a linear mapping:

3.1 Matrix Representation

- **2D Homogeneous Matrix (3×3):**

$$T(\theta, p_x, p_y) = \begin{bmatrix} \cos \theta & -\sin \theta & p_x \\ \sin \theta & \cos \theta & p_y \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} r_x \\ r_y \\ 1 \end{bmatrix} = T \begin{bmatrix} q_x \\ q_y \\ 1 \end{bmatrix}$$

- **3D Homogeneous Matrix (4×4):**

$$T = \begin{bmatrix} R & p \\ \mathbf{0}^T & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad Tq = Rq + p$$

- **Interpretation:** Rotation is applied first, followed by translation.
- **Key Properties:** Invertible, composable, generalizes pure rotation ($p = 0$) and pure translation ($\theta = 0$). T is *not* orthogonal.

4 Composition Rules & Ordering

- **Non-Commutativity:** Matrix multiplication generally does not commute ($T_1 T_0 \neq T_0 T_1$).
- **Absolute Transformations (Fixed Frame):** When transformations are specified relative to the **same** fixed world frame, **pre-multiply (left-multiply)** successive transformations:

$$C = B \cdot A$$

- **Relative Transformations (Moving Frame):** When transformation T_j^i transforms frame i to j (wrt i) and T_k^j transforms frame j to k (wrt j), **post-multiply (right-multiply)** successive relative transformations:

$$T_k^i = T_j^i \cdot T_k^j$$

5 Summary Cheat Sheet

Concept	2D Space	3D Space
Vector Dimension	$\mathbb{R}^2$	$\mathbb{R}^3$
Rotation Matrix	2×2 , single angle θ	3×3 , 3 principal axes (R_x, R_y, R_z)
Rotations Commute?	Yes (single axis)	No (across different axes)
Homogeneous Matrix	3×3	4×4
Composition Rule	Left-mult (Absolute) / Right-mult (Relative)	Left-mult (Absolute) / Right-mult (Relative)